Generalizing Score based Generative Model Applications

Anonymous Author(s)
Affiliation
Address
email

Abstract

Score-based generative models (SGMs) have achieved remarkable empirical success, yet their theoretical foundations remain incomplete. Two fundamental questions are largely unresolved: for which distributions can the backward SDE be rigorously justified, and under what conditions does the approximate diffusion converge to the target distribution? All existing theoretical analyses rely on strong assumptions on the data distribution, and the problem of heavy-tailed distributions remains particularly open. In this paper, we address both questions. We first prove that the backward SDE is well-posed for any data distribution, establishing the diffusion framework in full generality. We then show that combining early stopping with a flexible initialization scheme — replacing the standard Gaussian prior with a distribution adapted to the target — is sufficient to guarantee KL convergence of the approximate diffusion without any assumption on the target distribution, generalizing [Conforti et al., 2025, Theorem 1] to this end. On the practical side, we validate our initialization-aware approach on standard benchmarks, showing that a well-chosen initialization improves generation quality while reducing the number of required denoising steps. We further take a first step toward heavy-tailed distribution modeling with standard diffusion models, discussing suitable families of initialization priors and providing empirical evidence on toy, synthetic climatic, and real climatic datasets that proper initialization meaningfully improves over standard diffusion in this challenging regime.

1 Introduction

In recent years, generative models have emerged as a cornerstone of machine learning due to their capacity to sample high-dimensional distributions. Score-based generative models (SGMs) were introduced in their discrete form by Ho et al. [Ho et al., 2020] and in their continuous formulation by Song et al. [Song et al., 2021], building on several years of pioneering work [Sohl-Dickstein et al., 2015, Song and Ermon, 2019]. Since then, they have achieved state-of-the-art performance in image and video generation, surpassing GANs, VAEs, and Normalizing Flows [Dhariwal and Nichol, 2021, Müller-Franzes et al., 2022]. Their frameworks form the foundation of Stable Diffusion [Rombach et al., 2022] and many of the most prominent publicly available models today, including DALL·E 3 [Betker et al., 2023], Imagen 3 [Baldridge et al., 2024], and recent advanced architectures like FLUX [Black Forest Labs, 2025]. SGMs have also demonstrated competitive results in audio [Chen et al., 2021, Kong et al., 2021] and text generation [Li et al., 2022, Zou et al., 2023], positioning them as a flexible and powerful framework across diverse data modalities.

The core idea of SGMs is twofold: the data distribution is gradually corrupted by adding Gaussian noise, and a neural network is trained to reverse this process, recovering the original sample from its corrupted version. New samples are then generated by starting from standard Gaussian noise and iteratively applying the trained network to progressively denoise it.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Theoretical work on SGMs pursues a twofold goal: understanding *how* the approximate backward diffusion converges to the distribution of interest, and proving *that* it does. While it is clear that approximating the score function and using an approximate distribution both introduce errors in generation, it remains an open question under which conditions these approximations are sufficient to guarantee convergence to the true distribution. To establish such guarantees, existing works typically rely on strong assumptions on the data distribution. For instance, [Bortoli, 2022] restricts to bounded support, [Gao et al., 2025] requires log-concavity, and [Conforti et al., 2025] assumes a finite moment of order eight. More broadly, all existing theoretical analyses of SGMs impose some form of strong assumption on the data distribution [Bruno and Sabanis, 2025, Silveri and Ocello, 2025, Bortoli et al., 2021, Strasman et al., 2025].

A more fundamental issue concerns the backward SDE itself. The founding paper on continuous SGMs [Song et al., 2021] relies on [Anderson, 1982] to introduce it, yet [Anderson, 1982] establishes this result only for distributions with bounded support. While other works [Bortoli et al., 2021] invoke alternative references [Haussmann and Pardoux, 1986], it remains unclear what minimal assumptions on the data distribution are required to rigorously justify the backward SDE, and therefore the diffusion framework itself.

Understanding for which distributions the diffusion framework can be rigorously introduced, and when convergence can be established, is thus a question of both theoretical and practical importance. On the practical side, the recent surge in high-quality image generation has prompted researchers to explore whether diffusion can be extended to heavy-tailed distributions. Several approaches have been proposed, including normalizing flows [Hickling and Prangle, 2025, McDonald et al., 2022, Laszkiewicz et al., 2022], variational autoencoders [Lafon et al., 2023], GANs [Allouche et al., 2022, Huster et al., 2021], and diffusion-based methods [Pandey et al., 2025, Yoon et al., 2022, Shariatian et al., 2025], yet none has fully solved the problem. Several works [Pandey et al., 2025, Shariatian et al., 2025] explicitly highlight the difficulties SGMs face in this regime, but it remains unclear whether these failures stem from fundamental limitations of the framework or from practical issues such as training instability, poor initialization, or SDE discretization.

In this paper, we show that early stopping and proper initialization together allow the diffusion framework to be applied, theoretically, to any distribution. The idea of improving generation through non-standard initialization has already been explored [Zheng et al., 2023, Zand et al., 2024, Frey et al., 2024, gil Lee et al., 2022, Shen et al., 2025, Scholz and Turner, 2026], but always with the goal of improving efficiency or generation quality. Our contribution is fundamentally different: we provide a theoretical guarantee that diffusion models can be applied to *any* distribution in the sense of KL convergence, without any assumption on the latter.

Precisely, this paper makes the following contributions.

- We prove that the backward SDE is well-posed for any data distribution, establishing that the diffusion framework is theoretically grounded in full generality.
- We prove that combining early stopping with a flexible initialization scheme that replaces the standard Gaussian prior is sufficient to guarantee convergence of the approximate diffusion to the target distribution without any assumption on the latter, generalizing [Conforti et al., 2025, Theorem 1] to this end.
- We validate our initialization-aware approach on standard benchmarks, demonstrating that a well-chosen initialization improves generation quality while reducing the number of required denoising steps.
- We take a first step toward modeling heavy-tailed distributions with standard diffusion models. We discuss what families of distributions are suitable choices for the initialization prior, and provide empirical evidence on toy datasets, synthetic climatic data, and real climatic datasets that proper initialization meaningfully improves over standard diffusion in this regime.

2 Generalized Variance Exploding diffusion

We discuss in this section the existence of the forward and backward diffusion equations in the context of Variance Exploding framework. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathrm{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ the space of

square-integrable random variables on Ω. We denote by μ_{data} the unknown data distribution on $\mathbb{R}^d$, and the goal of SGMs is to generate new samples from μ_{data} given only an observed dataset.

2.1 Forward and Backward Process

Given $\overrightarrow{X}_0 \sim \mu_{\text{data}}$ the distribution of interest and the standard forward Variance Exploding (VE) equation [Song et al., 2021]

$$\mathrm{d}\overrightarrow{X}_t = g_t \, \mathrm{dB}_t \ . \tag{1}$$

where $g_t : \mathbb{R}^+ \to \mathbb{R}$ is assumed to be bounded and infinitely differentiable, we can state by applying Karatzas and Shreve [1991, Theorem 2.5] without any assumption on the data distribution that the forward solution $(\overrightarrow{X}_t)_{t\in[0,T]}$ is defined on all intervall for $T > 0$. Defining

$$\sigma_t^2 \coloneqq \int_0^t g_s \, \mathrm{d}s \ .$$

It can be stated that the solution

$$\overrightarrow{X}_t = \overrightarrow{X}_0 + \int_0^t g_s^2 \mathrm{d}s \overset{\mathcal{L}}{=} \overrightarrow{X}_0 + \sigma_t Z \ ,$$

Therefore as well known in the literature the solution simply results to be the convolution of the data distribution with a gaussian. In particular, for $t > 0$, $\overrightarrow{X}_t$ admits a density $\vec{p}_t(x)$ with respect to the Lebesgue measure $\lambda_{\mathbb{R}^d}$ and this density is highly regular. In particular the function $(t, x) \mapsto \vec{p}_t(x) \in C^\infty([\delta, T] \times \mathbb{R}^d)$, as shown in Lemma —- .

Concerning the backward equation the situation, might seem more complicated. The existence of the backward equation is not given for any distribution and any forward sde admitting an explicit solution. In fact the sufficient condition stated in reference articles [Anderson, 1982, Haussmann and Pardoux, 1986] require the data distribution or the density of the forward (when available) to respect some regularity conditions. Thankfully in the (VE) context where the forward solution is simply the convolution of the distributin with a gaussian, this doesn't represent a problem as stated in our first proposition.

Proposition 2.1. *Given μ_{data} a distribution and $\overrightarrow{X}_t, \vec{p}_t(x)$ the solutions of the* (1) *the backward equation*

$$\mathrm{d}\overleftarrow{X}_t = \bar{g}_t^2 \nabla \log \vec{p}_{T-t}(\overleftarrow{X}_t)\mathrm{d}t + \bar{g}_t \, \mathrm{d}\tilde{\mathrm{B}}_t \ , \tag{2}$$

where $\bar{g}_t \coloneqq g_{T-t}$ and $\tilde{\mathrm{B}}_t$ is an independent Brownian motion. For all initialization $\overleftarrow{X}_0$ and for all $\delta > 0$ (2) *admits a strong and unique solution $\overleftarrow{X} = \{\overleftarrow{X}_t\}_{[0,T-\delta]}$ on the interval $[0, T-\delta]$. In particular if $\overleftarrow{X}_0 \sim \vec{p}_T$, for all $\delta > 0$ it holds $\overleftarrow{X} \overset{\mathcal{L}}{=} \{\overrightarrow{X}\}_{[\delta,T]}$.*

The proposition that we presented establishes the existence of a backward strong solution for all

2.2 KL control on approximated process

In practice, μ_* is not available; instead, we only have access to an i.i.d. sample $\overrightarrow{X}_0^i \sim \mu_*$, $1 \le i \le n$. Consequently, the solution of Equation (2) that would enable sample generation is not directly accessible, since neither the score nor the terminal distribution $\vec{p}_T$ of the backward process are known. Furthermore, the backward SDE does not admit a closed-form solution, making it necessary to rely on discretization schemes.

For the purpose of theoretical analysis, following common practice, we may rely on either the Euler–Maruyama scheme Gao et al. [2025], Silveri and Ocello [2025] or the Exponential Euler–Maruyama scheme Conforti et al. [2025], Strasman et al. [2025]. Remarkably, in the VE setting these two schemes coincide, allowing for a unified treatment of the KL bound.

Score approximation and discretization. By replacing the real score function $\nabla \log \vec{p}_{T-t}$ using a trained neural network $s_\theta(T-t, \cdot)$, we introduce the discretized backward dynamics

$$\mathrm{d}X_t^\theta = \bar{g}_t^2 s_\theta(T - t_k, X_{t_k}^\theta) \, \mathrm{d}t + \bar{g}_t \, \mathrm{d}\tilde{\mathrm{B}}_t \ , \tag{3}$$

for $t \in [t_k, t_{k+1}]$, where $t_{k+1} = t_k + h_{k+1}$, with $t_0 = 0$ and $t_N = T - \delta$ which can be equivalently written, for $t \in [0, T-\delta]$, as

$$\mathrm{d}X_t^\theta = \sum_{k=0}^{N-1} \bar{g}_t^2 s_\theta(T - t_k, X_{t_k}^\theta) 1_{t \in [t_k, t_{k+1}]} \, \mathrm{d}t + \bar{g}_t \, \mathrm{d}\tilde{\mathrm{B}}_t \ . \tag{4}$$

This discretization scheme results in the following stochastic sampling dynamics:

$$X_{t_{k+1}}^\theta = X_{t_k}^\theta + (\sigma_{T-t_k}^2 - \sigma_{T-t_{k+1}}^2) s_\theta(T - t_k, X_{t_k}^\theta) + \sqrt{\sigma_{T-t_k}^2 - \sigma_{T-t_{k+1}}^2} Z \ , \tag{5}$$

for $k = 0, ..., N-1$, $Z \sim \mathcal{N}(0, \mathrm{I}_d)$. We initialize the backward not with the gaussian standaard distribution but using a general initialization p_0^θ such that p_0^θ and $\lambda_{\mathbb{R}^d}$ are mutually absolutely continuous. We will use p_0^θ to point to the law of X_0^θ and to its density, the context will clarify what we mean. We assume that

$$\mathrm{D}_{\mathrm{KL}} \left(\vec{p}_t || p_0^\theta \right) < \infty \ .$$

Furthermore we make the following important assumption of score approximation

H1 There exists $\varepsilon > 0$ such that for all $0 \leq k \leq N-1$,

$$\mathbb{E}\Big[\| \nabla \log \vec{p}_{T-t_k}(\overleftarrow{X}_{t_k}) - s_\theta(T - t_k, \overleftarrow{X}_{t_k}) \|^2 \Big] \leq \varepsilon \ .$$

Consider the following additional notations: for all θ, $t \in [0, T]$, $0 \leq k \leq N-1$,

$$\begin{aligned} \mathcal{E}_\theta(t_k, \overleftarrow{X}_{t_k}) &= \mathsf{s}_{T-t_k}(\overleftarrow{X}_{t_k}) - s_\theta(T - t_k, \overleftarrow{X}_{t_k}) \ , \\ \mathcal{E}(t_k, t, \overleftarrow{X}_{t_k}, \overleftarrow{X}_t) &= \mathsf{s}_{T-t_k}(\overleftarrow{X}_{t_k}) - \mathsf{s}_{T-t}(\overleftarrow{X}_t) \ . \end{aligned} \tag{6}$$

Theorem 2.2. *For all distribution μ_{data}, for all $\delta > 0$, it holds that*

$$\mathrm{D}_{\mathrm{KL}} \left(\vec{p}_\delta || p_{T-\delta}^\theta \right) \leq \mathsf{E}_{\mathrm{init}}(\theta) + \mathsf{E}_{\mathrm{train}}(\theta) + \mathsf{E}_{\mathrm{disc}} \ ,$$

with

$$\mathsf{E}_{\mathrm{init}}(\theta) = \mathrm{D}_{\mathrm{KL}} \left(\vec{p}_T || p_0^\theta \right) \ , \tag{7}$$

$$\mathsf{E}_{\mathrm{train}}(\theta) = \sum_{k=0}^{N-1} \gamma_k \mathbb{E} \left[\left\| \mathcal{E}_\theta(t_k, \overleftarrow{X}_{t_k}) \right\|^2 \right] \ , \tag{8}$$

$$\mathsf{E}_{\mathrm{disc}} = \sum_{k=0}^{N-1} \int_{t_k}^{t_{k+1}} \bar{g}_t^2 \mathbb{E} \left[\left\| \mathcal{E}(t_k, t, \overleftarrow{X}_{t_k}, \overleftarrow{X}_t) \right\|^2 \right] \mathrm{d}t \ , \tag{9}$$

where $\mathcal{E}_\theta(t_k, \overleftarrow{X}_{t_k})$ and $\mathcal{E}(t_k, t, \overleftarrow{X}_{t_k}, \overleftarrow{X}_t)$ are defined in (6)*. In addition, the term $\mathsf{E}_{\mathrm{disc}}$ is upper bounded by*

$$\mathsf{E}'_{\mathrm{disc}} = \max_{k=0,\dots,N-1} \gamma_k \left\{ \mathcal{I}(\vec{p}_\delta) - \mathcal{I}(\vec{p}_T) \right\} \ . \tag{10}$$

Assume in addition that H1 holds. Then,

$$\mathrm{D}_{\mathrm{KL}} \left(\vec{p}_\delta || p_{T-\delta}^\theta \right) \leq \mathrm{D}_{\mathrm{KL}} \left(\vec{p}_T || p_0^\theta \right) + \sigma_T^2 \epsilon + \mathsf{E}'_{\mathrm{disc}} \ . \tag{11}$$

Therefore the gaussian initialization doesnt work in many context, for example if $\overrightarrow{X}_0$ is a Frechet and p_0^θ a gaussian the $\mathrm{D}_{\mathrm{KL}}(||)$ will be infinite ant therefore there will be no bound. But by extending to general initialization we cover all distributions.

3 Experiments

- We present experiments with normal datasets. We show that everything works well and we also saved some time. This is not new so this part will not be central.
- We present experiments with heavytail distribution. We show that it works pretty well, so that there is hope and ongoing work to do to correctly choose the flow p_θ which is exactly the point of research where we are stuck.
- We can include Nicolas and he could help me with doing the simulations on the heavytail datasets. As datasets i propose the synthetic dataset i created and the Comephere data which can be cleaned by me and Nicolas.

References

Michaël Allouche, Stéphane Girard, and Emmanuel Gobet. Ev-gan: Simulation of extreme events with relu neural networks. *Journal of Machine Learning Research*, 23(120):1–39, 2022. URL http://jmlr.org/papers/v23/21-0663.html.

Brian DO Anderson. Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326, 1982.

Jason Baldridge, Jakob Bauer, Mukul Bhutani, Nicole Brichtova, Andrew Bunner, Lluis Castrejon, Kelvin Chan, Yichang Chen, Sander Dieleman, Yuqing Du, Zach Eaton-Rosen, Hongliang Fei, Nando de Freitas, Yilin Gao, Evgeny Gladchenko, Sergio Gómez Colmenarejo, Mandy Guo, Alex Haig, Will Hawkins, Hexiang Hu, Huilian Huang, Tobenna Peter Igwe, Christos Kaplanis, Siavash Khodadadeh, Yelin Kim, Ksenia Konyushkova, Karol Langner, Eric Lau, Rory Lawton, Shixin Luo, Soňa Mokrá, Henna Nandwani, Yasumasa Onoe, Aäron van den Oord, Zarana Parekh, Jordi Pont-Tuset, Hang Qi, Rui Qian, Deepak Ramachandran, Poorva Rane, Abdullah Rashwan, Ali Razavi, Robert Riachi, Hansa Srinivasan, Srivatsan Srinivasan, Robin Strudel, Benigno Uria, Oliver Wang, Su Wang, Austin Waters, Chris Wolff, Auriel Wright, Zhisheng Xiao, Hao Xiong, Keyang Xu, Marc van Zee, Junlin Zhang, Katie Zhang, Wenlei Zhou, Konrad Zolna, and . . . . Imagen 3. *arXiv preprint*, arXiv:2408.07009, 2024. doi: 10.48550/arXiv.2408.07009. URL https://arxiv.org/abs/2408.07009.

James Betker, Gabriel Goh, Li Jing, Tim Brooks, Jianfeng Wang, Linjie Li, Long Ouyang, Juntang Zhuang, Joyce Lee, Yufei Guo, Wesam Manassra, Prafulla Dhariwal, Casey Chu, Yunxin Jiao, and Aditya Ramesh. Improving image generation with better captions. Technical report, OpenAI and Microsoft, 2023. DALL-E 3 System Card.

Black Forest Labs. Flux.1-dev: 12b-parameter rectified flow transformer for text-to-image generation. https://huggingface.co/black-forest-labs/FLUX.1-dev, 2025.

Valentin De Bortoli. Convergence of denoising diffusion models under the manifold hypothesis. *Transactions on Machine Learning Research*, 2022. ISSN 2835-8856. URL https://openreview.net/forum?id=MhK5aXo3gB. Expert Certification.

Valentin De Bortoli, James Thornton, Jeremy Heng, and Arnaud Doucet. Diffusion schrödinger bridge with applications to score-based generative modeling. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021. URL https://openreview.net/forum?id=9BnCwiXB0ty.

Stefano Bruno and Sotirios Sabanis. Wasserstein convergence of score-based generative models under semiconvexity and discontinuous gradients. *Transactions on Machine Learning Research*, 09 2025. URL https://openreview.net/forum?id=vS9iVRB7XF. Published 09/2025.

Nanxin Chen, Yu Zhang, Heiga Zen, Ron J Weiss, Mohammad Norouzi, and William Chan. Wavegrad: Estimating gradients for waveform generation. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=NsMLjcFaO8O.

Giovanni Conforti, Alain Durmus, and Marta Gentiloni Silveri. Kl convergence guarantees for score diffusion models under minimal data assumptions. *SIAM Journal on Mathematical Analysis*, 7(1): 86–109, 2025. doi: 10.1137/23M1613670. URL https://doi.org/10.1137/23M1613670.

Prafulla Dhariwal and Alexander Quinn Nichol. Diffusion models beat GANs on image synthesis. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021. URL https://openreview.net/forum?id=AAWuCvzaVt.

Nathan C. Frey, Dan Berenberg, Karina Zadorozhny, Joseph Kleinhenz, Julien Lafrance-Vanasse, Isidro Hotzel, Yan Wu, Stephen Ra, Richard Bonneau, Kyunghyun Cho, Andreas Loukas, Vladimir Gligorijevic, and Saeed Saremi. Protein discovery with discrete walk-jump sampling. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=zMPHKOmQNb.

Xuefeng Gao, Hoang M. Nguyen, and Lingjiong Zhu. Wasserstein convergence guarantees for a general class of score-based generative models. *Journal of Machine Learning Research*, 26(43): 1–54, 2025. URL http://jmlr.org/papers/v26/24-0902.html.

Sang gil Lee, Heeseung Kim, Chaehun Shin, Xu Tan, Chang Liu, Qi Meng, Tao Qin, Wei Chen, Sungroh Yoon, and Tie-Yan Liu. Priorgrad: Improving conditional denoising diffusion models with data-dependent adaptive prior. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=_BNiN4IjC5.

U. G. Haussmann and E. Pardoux. Time reversal of diffusions. *Annals of Probability*, 14(4): 1188–1205, October 1986. ISSN 0091-1798, 2168-894X. doi: 10.1214/aop/1176992362.

Tennessee Hickling and Dennis Prangle. Flexible tails for normalizing flows. In *Forty-second International Conference on Machine Learning*, 2025. URL https://openreview.net/forum?id=Z6RsbHAJk5.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 6840–6851. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html.

Todd Huster, Jeremy E.J. Cohen, Zinan Lin, Kevin Chan, Charles Kamhoua, Nandi Leslie, Cho-Yu Jason Chiang, and Vyas Sekar. Pareto gan: Extending the representational power of gans to heavy-tailed distributions. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *PMLR*, pages 4690–4699. PMLR, 2021. URL http://proceedings.mlr.press/v139/huster21a.html.

Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*, volume 113 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 2 edition, 1991. doi: 10.1007/978-1-4612-0949-2.

Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. Diffwave: A versatile diffusion model for audio synthesis. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=a-xFK8Ymz5J.

Nicolas Lafon, Philippe Naveau, and Ronan Fablet. A vae approach to sample multivariate extremes. *arXiv preprint arXiv:2306.10987*, 2023.

Mike Laszkiewicz, Johannes Lederer, and Asja Fischer. Marginal tail-adaptive normalizing flows, 2022. URL https://openreview.net/forum?id=per0G3dnkYh.

Xiang Lisa Li, John Thickstun, Ishaan Gulrajani, Percy Liang, and Tatsunori B. Hashimoto. Diffusion-lm improves controllable text generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 4328–4343, 2022. URL https://github.com/XiangLi1999/Diffusion-LM.

Andrew McDonald, Pang-Ning Tan, and Lifeng Luo. Comet flows: Towards generative modeling of multivariate extremes and tail dependence. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI)*, pages 3328–3334. IJCAI Organization, 2022.

Gustav Müller-Franzes, Jan Moritz Niehues, Firas Khader, . . . , and Daniel Truhn. Diffusion probabilistic models beat gans on medical images. *arXiv preprint arXiv:2212.07501*, 2022. URL https://arxiv.org/abs/2212.07501.

Kushagra Pandey, Jaideep Pathak, Yilun Xu, Stephan Mandt, Michael Pritchard, Arash Vahdat, and Morteza Mardani. Heavy-tailed diffusion models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=tozlOEN4qp.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. High-Resolution Image Synthesis with Latent Diffusion Models . In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10674–10685, Los Alamitos, CA, USA, June 2022. IEEE Computer Society. doi: 10.1109/CVPR52688.2022.01042. URL https://doi.ieeecomputersociety.org/10.1109/CVPR52688.2022.01042.

Jonas Scholz and Richard E. Turner. Warm starts accelerate conditional diffusion, 2026. URL https://openreview.net/forum?id=3kzDWrohSY.

Dario Shariatian, Umut Simsekli, and Alain Oliviero Durmus. Denoising levy probabilistic models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=SYmUS6qRub.

Yirong Shen, Lu GAN, and Cong Ling. Information theoretic learning for diffusion models with warm start. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=3IbKbmNci3.

Marta Gentiloni Silveri and Antonio Ocello. Beyond log-concavity and score regularity: Improved convergence bounds for score-based generative models in w2-distance. In *Forty-second International Conference on Machine Learning*, 2025. URL https://openreview.net/forum?id=gVenpttGec.

Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2256–2265, Lille, France, 07–09 Jul 2015. PMLR. URL https://proceedings.mlr.press/v37/sohl-dickstein15.html.

Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. URL https://openreview.net/forum?id=B1lcYrBgLH.

Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=PxTIG12RRHS.

Stanislas Strasman, Antonio Ocello, Claire Boyer, Sylvain Le Corff, and Vincent Lemaire. An analysis of the noise schedule for score-based generative models. *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856. URL https://openreview.net/forum?id=BlYIPa0Fx1.

Eunbi Yoon, Keehun Park, Jinhyeok Kim, and Sungbin Lim. Score-based generative models with lévy processes. In *NeurIPS 2022 Workshop on Score-Based Methods*, 2022. URL https://openreview.net/forum?id=ErzyBArv6Ue.

Mohsen Zand, Ali Etemad, and Michael Greenspan. Diffusion models with deterministic normalizing flow priors. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL https://openreview.net/forum?id=ACMNVwcR6v.

Huangjie Zheng, Pengcheng He, Weizhu Chen, and Mingyuan Zhou. Truncated diffusion probabilistic models and diffusion-based adversarial auto-encoders. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=HDxgaKk956l.

Hao Zou, Zae Myung Kim, and Dongyeop Kang. A survey of diffusion models in natural language processing. *arXiv preprint arXiv:2305.14671*, 2023. URL https://arxiv.org/abs/2305.14671.